METRIC

FF-B-171/15

30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, PRIMARILY RADIAL, COUNTERBORED OUTER RING, CONTACT ANGLE UP TO AND INCLUDING 10°, NON-SEPARABLE, ANSI/AFBMA DIMENSION SERIES 19.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

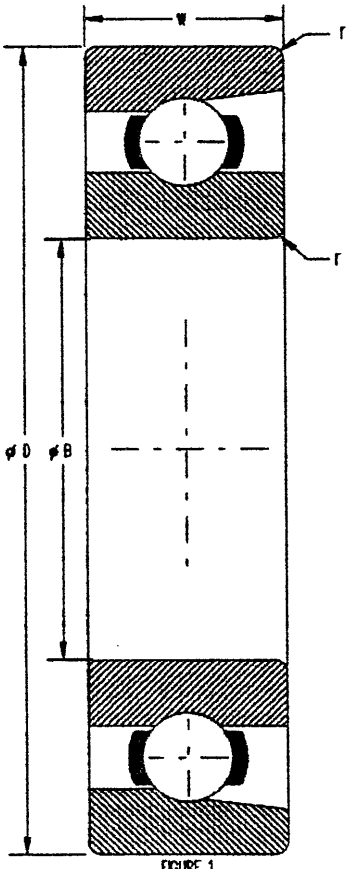


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

BEARING CONFIGURATIONS

AMSC/NA FSC 3110

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size code in table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and Seals All primarily radial bearings with counterbored outer rings shall be open (no shields or seals, see figure 1) and designated by the code "A" in the part number.

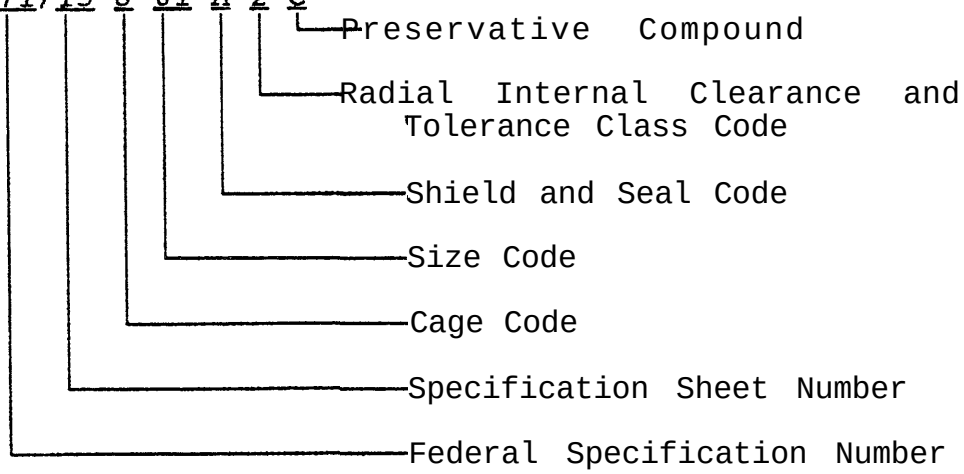
4. Radial Internal Clearance. The tolerance limits fur radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

6. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code "C" in the part number.

7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and seals, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE : FFB171/15 J 01 A 2 C



FFB171/15J01A2C indicates - Bearing, cage-pressed steel or molded plastic; bore 12mm; outside diameter 24mm; width 6mm; open (no shields or seals); radial internal clearance symbol 0 (normal); tolerance class ABEC-1; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic Load Rating (Pounds)
00	10	22	6	0.3	12	510
01	12	24	6	0.3	14	540
02	15	28	7	0.3	17	805
03	17	30	7	0.3	19	845
04	20	37	9	0.3	22	1560
05	25	42	9	0.3	27	1270
06	30	47	9	0.3	32	1580
07	35	55	10	0.6	39	1850
08	40	62	12	0.6	44	2355
09	45	68	12	0.6	49	2480
10	50	72	12	0.6	54	3190
11	55	80	13	1.0	60	3740
12	60	85	13	1.0	65	3930
13	65	90	13	1.0	70	3995
14	70	100	16	1.0	75	5480
15	75	105	16	1.0	80	5580
16	80	110	16	1.0	85	5680
17	85	120	18	1.1	91.5	7405
18	90	125	18	1.1	96.5	8665
19	95	130	18	1.1	101.5	7680
20	100	140	20	1.1	106.5	9640
21	105	145	20	1.1	111.5	9820
22	110	150	20	1.1	116.5	9005
24	120	165	22	1.1	126.5	12110
26	130	180	24	1.5	138	14705
28	140	190	24	1.5	148	14895
30	150	210	28	2.0	159	20555
32	160	220	28	2.0	169	20860
34	170	230	28	2.0	179	21680
36	180	250	33	2.0	189	27710
38	190	260	33	2.0	199	28150
40	200	280	38	2.1	211	35410

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

TABLE III. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (Less than normal)	ABEC-1
2	Symbol 0 (Normal)	
3	Symbol 3 (Greater than normal)	
4	Symbol 4 (Greater than symbol 3)	
5	Symbol 2 (Less than normal)	ABEC-5
6	Symbol 0 (Normal)	
7	Symbol 3 (Greater than normal)	
8	Symbol 4 (Greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-15)

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